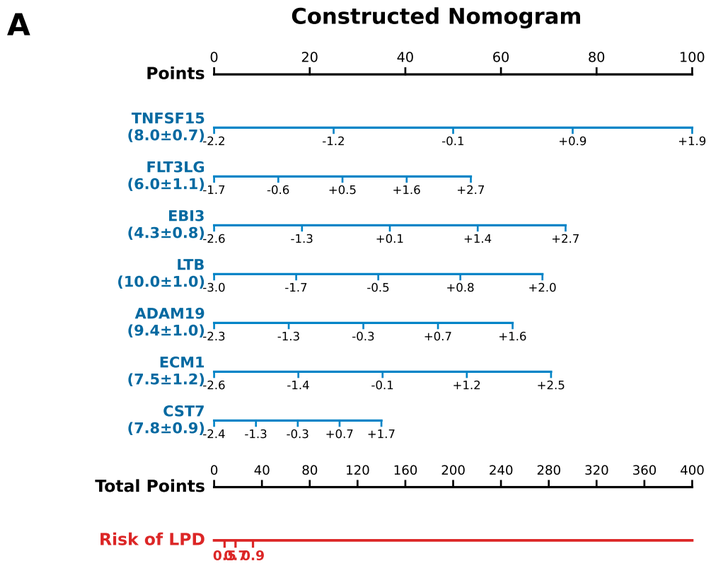
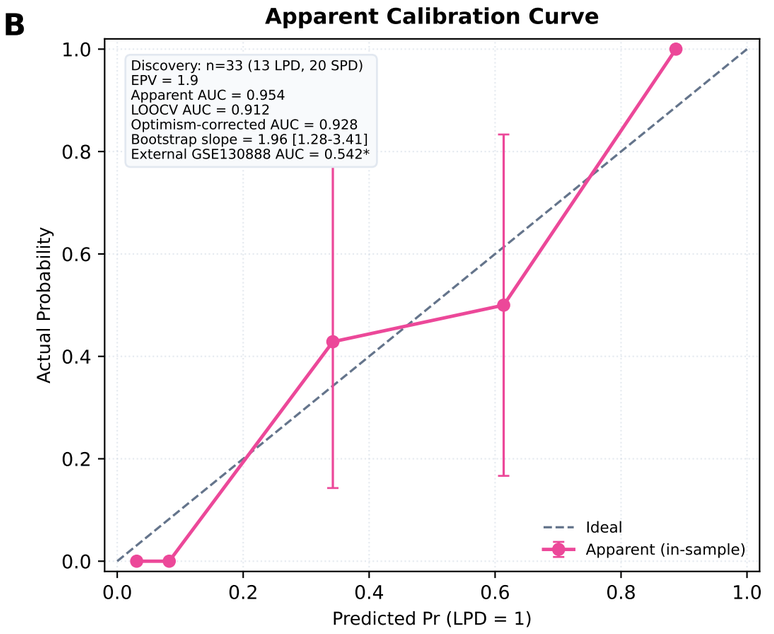
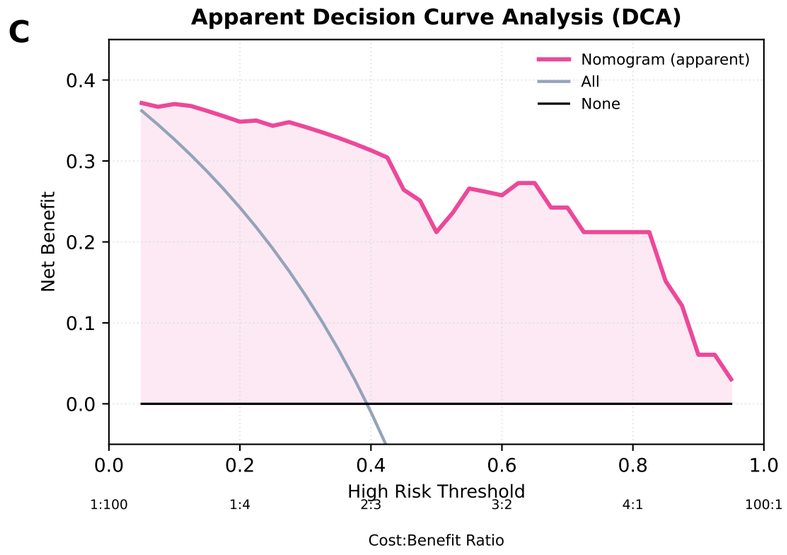
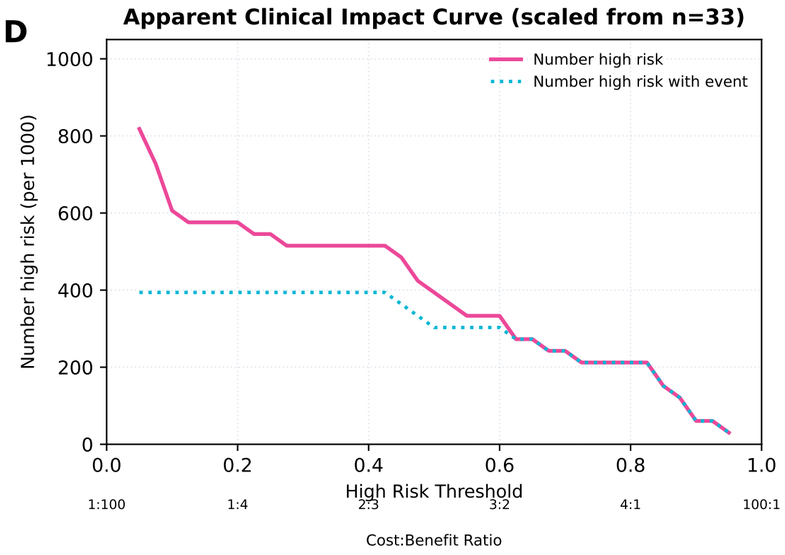
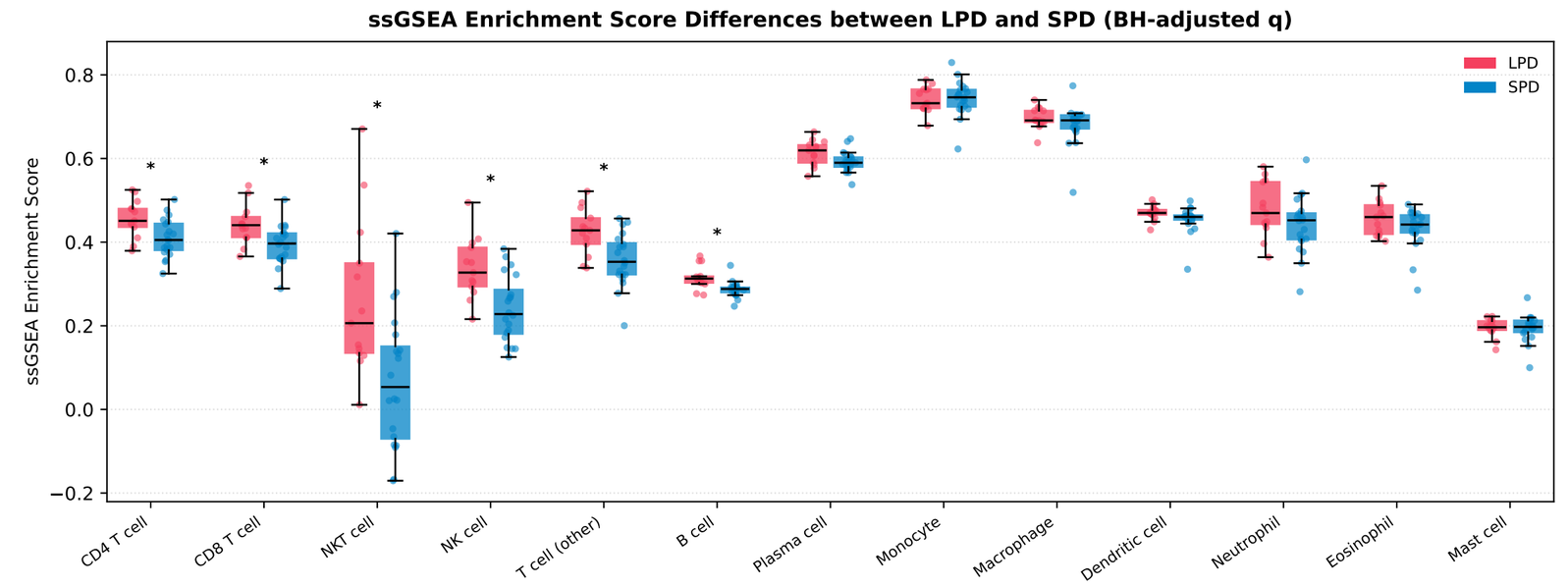










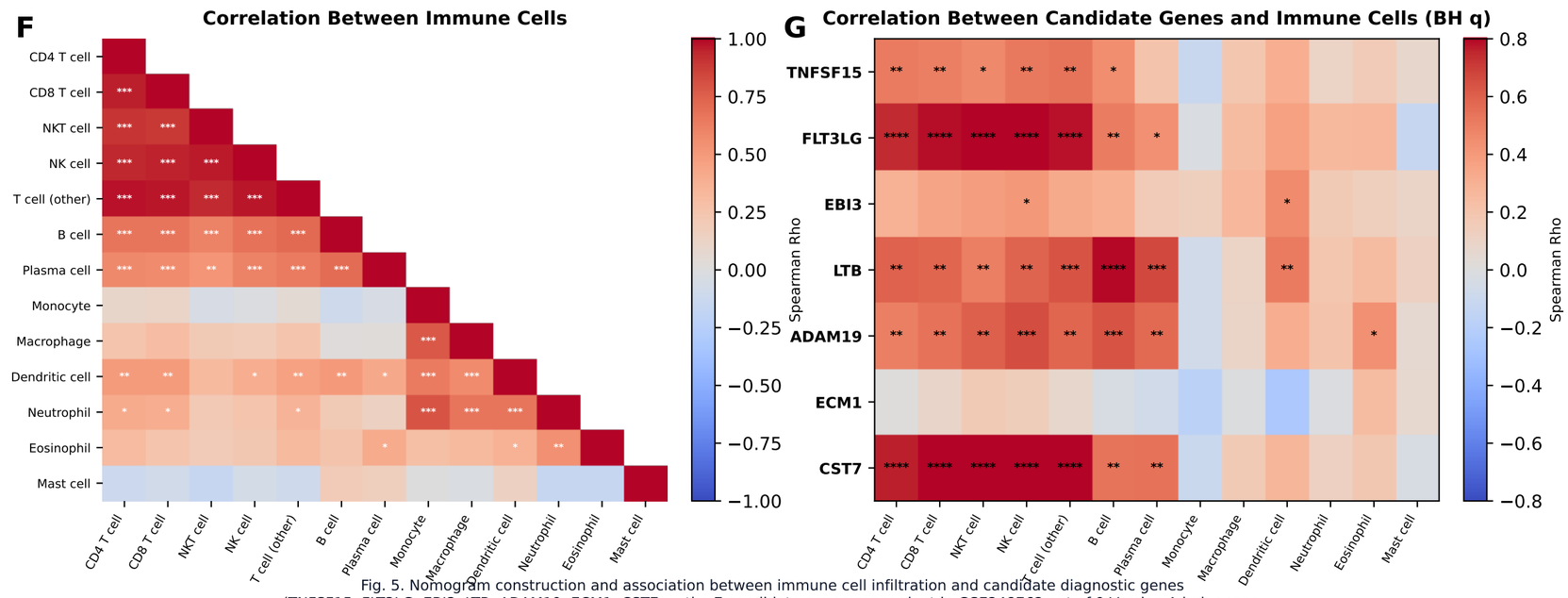


Fig. 5. Nomogram construction and association between immune cell infiltration and candidate diagnostic genes (TNFSF15, FLT3LG, EBI3, LTB, ADAM19, ECM1, CST7 — the 7 candidate genes concordant in GSE248762 out of 9 Version A hub genes; NAT6 was detected [all-cell AUC = 0.500, P = 1.0] and CCBE1 was detected and inverted [AUC = 0.067, P = 0.003], and both were dropped after viewing verification data). (A) Constructed nomogram (7-gene ridge model; gene labels display discovery mean ± SD). (B) Apparent (in-sample) calibration curve (5 equal-count bins with exact binomial CIs; bootstrap calibration slope = 1.96 [95% CI 1.28 –3.41]; *external frozen model AUC on GSE130888 = 0.542, subject to platform/scale differences). (C) Apparent DCA evaluating clinical net benefit vs treat-all/treat-none. (D) Apparent clinical impact curve (scaled to 1,000 from discovery n = 33). (E) ssGSEA enrichment score differences between LPD and SPD (6 of 13 cell types survive BH adjustment at q < 0.05: CD4 T cell, CD8 T cell, NKT cell, NK cell, T cell other, B cell; plasma cell q = 0.079, dendritic cell q = 0.069). (F) Spearman correlation between immune cell types. (G) Spearman correlation between 7 candidate genes and immune cell scores (38 of 91 tests survive BH FDR adjustment across the panel; note that 6 of 7 candidate genes appear as markers in CellMarker signatures, e.g. CST7 and LTB in lymphocyte sets). All panels: apparent (in-sample) discovery estimates; n = 33 (13 LPD, 20 SPD); 7 predictors. *q < 0.05, **q < 0.01, ***q < 0.001, ****q < 0.0001 (Benjamini –Hochberg adjusted).